Sparse Eisenstein radix family: screen, proof boundary, and paired costs

Question. Across $\pi_k=(2^k-1)+2^k\tau$, how do finite-fixture recoding geometry, one-scalar cold cost, and prepared-table throughput vary for $k=5,6,7$?

Scope and input. The experiment uses SEC 2 secp256k1, $E: y^2=x^3+7$ over $p=2^{256}-2^{32}-977$, and prime subgroup order $n=$FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141. The 12,288-scalar fixture SHA-256 is a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3. The curve catalog on origin/main has no secp256k1 EC1 record; this report identifies the exact standard curve and subgroup parameters and does not mint a new ID.

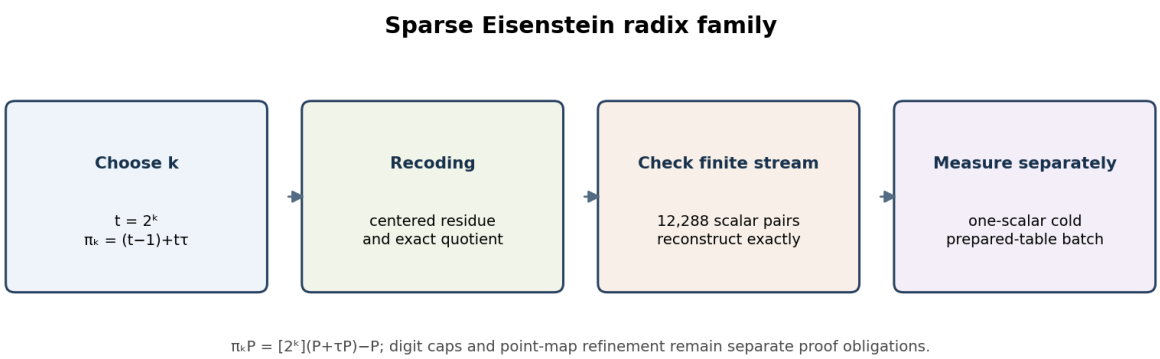


Figure 1: Sparse radix family work and evidence stages

Algebra and proof status. Set $t=2^k$, so $\pi_k=(t-1)+t\tau$, with $\tau^2=3\tau-3$. Pair multiplication is $((t-1)u-3tv, tu+(4t-1)v)$, its norm is $7t^2-5t+1$, and the point action is $[t](P+\tau P)-P$. The pinned SparseRadixFamily.lean proves those identities for every integer t, the quotient residue $\tau=4-7t$, its root polynomial, and norm scaling. It also proves terminal norm bounds for $k=5$ and $k=6$ under explicit per-step recurrence and digit-cap hypotheses. The Lean checker and committed receipt record successful Lean 4.34.1 builds and #print axioms; the source hash in the receipt matches the current file. I could not rerun Lean in this worktree because no lean executable is installed.

The recurrence hypotheses are not proved for the selected native digits. The integer screen and its committed receipt check exact reconstruction on the fixture, not a global termination theorem. This review used the committed screen receipt and source; it did not regenerate that Python-produced receipt. The recoding report says native_point_replay: pending for each row because this screen operates on integer pairs. Separate native panels below check the implemented point outputs for their frozen workloads; neither establishes a curve-wide formal refinement.

k	Radix	Norm	Digit seeds incl. zero	Fixture steps	Largest observed terminal norm
5	31+32τ	7,009	1,169	20	1
6	63+64τ	28,353	4,727	17	9
7	127+128τ	114,049	19,009	15	7

The counts and terminal values are finite-fixture observations. They do not prove selector optimality or a uniform digit cap. The chosen counts decrease as k rises in this fixture while the number of steps falls; the table geometry and measured costs below are the actual comparison evidence.

Native panels. Four retrospective Tier 1 comparisons ran serially on RunPod thyvmdqmzsz5ik, CPU 40; each had 16 pairs (96 arms, 48 pairs) and passing artifact/A/A gates. Cold panels charge table setup plus one scalar; prepared panels time 12,288 scalars after setup. Re-running each existing auditor reproduced its committed JSON receipt byte-for-byte.

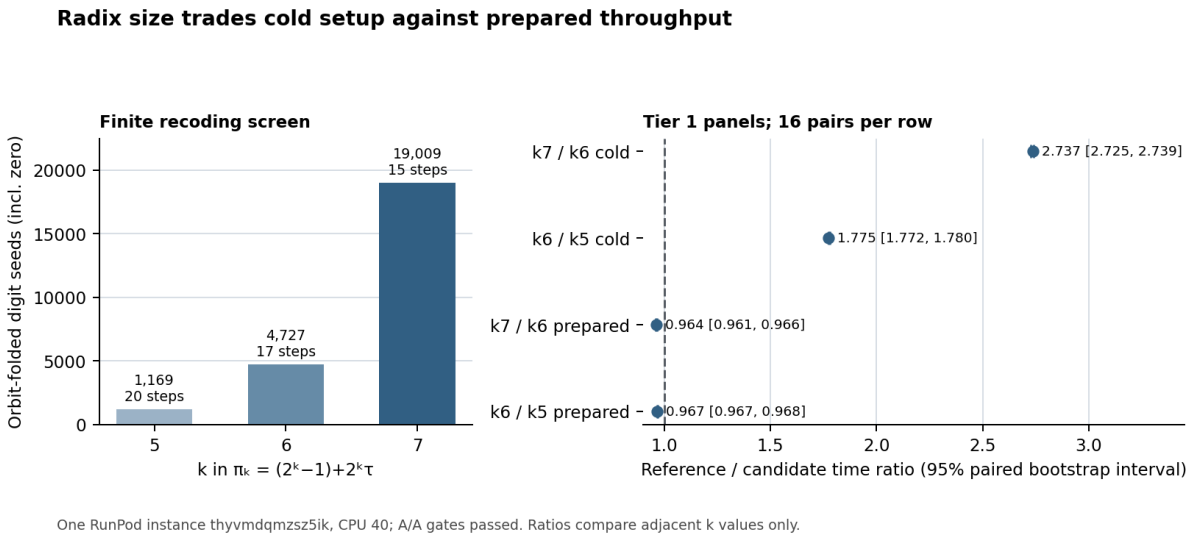


Figure 2: Finite recoding screen and paired Tier 1 ratios

Workload and comparison	Reference mean	Candidate mean	Reference/candidate paired median (95% interval)	Retained bytes, ref → cand
One-scalar cold, k=7 → k=6	255.343752 ms	93.577572 ms	2.737020 [2.725245, 2.738990]	2,636,452 → 922,596
One-scalar cold, k=6 → k=5	93.497948 ms	52.673690 ms	1.774959 [1.771918, 1.779729]	922,596 → 495,652
Prepared 12,288-point stream, k=7 → k=6	602.053380 ms	624.655146 ms	0.963705 [0.961166, 0.965874]	2,636,452 → 922,596
Prepared 12,288-point stream, k=6 → k=5	625.229301 ms	645.932010 ms	0.967429 [0.966595, 0.968334]	922,596 → 495,652

The panel jobs, in table order, are cold k7/k6 `1ebc1b95247c45bb90c957bed33ba2b1`, cold k6/k5 `8e13db8a4f57421f8bfe` prepared k7/k6 `664fc7a36b064850ae1c103b186f057a`, and prepared k6/k5 `6348a3b2fdc74dbf81fd861372894c54`. The linked study note indexes manifests, rules, raw archives, and source provenance. Cold panels favor the smaller candidate radix for setup plus one scalar; prepared panels favor the larger reference radix for this stream. These are separate workloads and adjacent comparisons, not one universal winner.

Native and graph limits. The k=6 overlay, k=5 overlay, and Tier 1 audits provide native point-replay evidence for the frozen workloads; the audit receipts check outputs, hashes, timing boundaries, and paired gates. The family screen remains pair-only (`native_point_replay: pending`). Uniform selected-digit caps for `k=5,6`, per-step code-to-Lean refinement, and the curve-wide point-map/group-law bridge remain open, so general correctness for every subgroup scalar is not established.

No new curve or isogeny route was found. The study tree has no canonical curve/isogeny graph to update; `origin/main` has no secp256k1 catalog record. The method and measurement graphs document this existing radix study and change no curve/map edge. The editable figure source reads committed receipts; SVG/PNG and this report's PDF are included.